

Nathaniel “Natey” Kim

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RESEARCH INTERESTS

My research focuses on developing human–robot interfaces that directly support how people move, interact, and function in daily life, particularly through assistive and rehabilitative robotic systems that operate as intuitive extensions and partners in human activity. My interests span physical and cognitive human–robot interaction, with a focus on human movement analysis, upper-limb exoskeletons, musculoskeletal modeling, human-in-the-loop control, and social robotics.

EDUCATION

University of Pennsylvania

May 2026

- **M.S.E. in Robotics** (submatriculation), GPA: **4.0/4.0**

Master’s Thesis: *Musculoskeletal Digital Twin Model for Real-Time Human-in-the-Loop Feedback Control Systems*

Advisor: Dr. Nadia Figueroa

- **B.S.E. in Bioengineering**, GPA: **4.0/4.0**

RESEARCH EXPERIENCE

University of Pennsylvania GRASP Laboratory — Research Engineer

Jun 2026 – Present

Graduate Research Assistant

Aug 2023 – May 2026

Hybrid Simulation- and EMG-Controlled Upper-Limb Exoskeleton

Jun 2025 – Present

- Developed a MyoSuite musculoskeletal digital twin using inverse dynamics, PD control, and quadratic programming, reproducing measured elbow kinematics with a mean tracking RMSE of 1.866° .
- Developed a causal temporal convolutional network that predicted subject-specific biceps and triceps EMG for unseen controller configurations with mean NRMSEs of 10.7% and 10.5%, respectively, at real-time latency.
- Integrated surface EMG, OptiTrack motion capture, motor torque, and exoskeleton encoder data across three participants, four external loads, and five controller configurations.

EMG-Based Achilles Tendinopathy Classification

Aug 2023 – Present

- Developed a real-time high-density EMG collection pipeline using custom 3D-printed hardware and laser-cut MMX/silicon fabric electrodes for movement and injury analysis.
- Built a 3D variational autoencoder to quantify Achilles tendinopathy severity, assess injury laterality, and classify rehabilitation exercises from high-density EMG.
- Detected Achilles tendinopathy in an average of 20 of 21 held-out evaluations and correctly classified healthy, unilateral, and bilateral pathology in 18 of 21.

PROFESSIONAL EXPERIENCE

Maingen (Y Combinator) — Resident Researcher (part-time)

Jul 2026 – Present

- Research machine-learning methods for generating and optimizing robotic end-effector CAD geometries using simulation-derived performance requirements.
- Develop simulation and evaluation pipelines for assessing generated designs through geometric validity, mechanical performance, and task-specific engineering constraints.

Pallas Robotics — Founding Mechanical Engineer

Aug 2025 – Present

- Translate Ember from an academic social-robotics research platform into an early-stage product for rehabilitative patient support and naturalistic human–robot interaction.
- Designed the mechanical enclosure and embedded architecture, integrating wireless communication, audio I/O, vision, display, power, and motor-control hardware.
- Developed LLM-based interaction logic for reminiscence dialogue and emergency guidance and authored human-subject and non-human-subject IRB submissions supporting studies of Ember.

PUBLICATIONS & MANUSCRIPTS

Sanghyub Lee, **Nathaniel Kim**, Dhyey Shah, Jason Chen, and Nadia Figueroa. “Towards Physiologically Informed Evaluation of Exoskeleton Controllers: A Simulation-Based Deep Learning Approach for Muscle Activity Prediction” Under review at *2026 IEEE-RAS International Conference on Humanoid Robots (Humanoids)*.

Maggie Wagner, Sanghyub Lee, **Nathaniel Kim**, Michael North, Nadia Figueroa, Josh Baxter, and Flavia Vitale. “Latent HD-sEMG Representation Learning for Achilles Tendinopathy Assessment.” Manuscript in preparation for submission to *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 2026.

RESEARCH & DESIGN PROJECTS

Verdigris: Assistive Ultrasound Guidance Device for Lumbar Puncture (BSE Senior Design) Aug 2025 – May 2026

- Co-developed an assistive medical device using ultrasound-based bone detection and real-time angle feedback to guide clinicians toward viable lumbar puncture needle trajectories.
- Designed the device enclosure and visual interface, integrated the ultrasound transducer, motor, needle-angle sensor, and control electronics, and conducted acoustic signal acquisition testing on tissue and spinal phantoms.
- Validated angle sensing and LED feedback to within 1° across controlled targets; contributed to a benchtop prototype achieving 80% optimal needle-path detection and 95% bone-detection accuracy.

Rehabilitative Driving Simulator

Jan 2023 – May 2023

- Developed a C#-based rehabilitative driving simulator for pediatric patients with hemiplegia in collaboration with engineers and clinicians in Jamaica.
- Designed and built custom steering and pedal interfaces, including sensor and embedded control circuitry, to integrate physical user inputs with the simulator.

PRESENTATIONS & POSTERS

Penn Undergraduate Research Mentoring Program Symposium — Presenter

2024

- Presented “From Muscle Signals to Motion: A Deep Learning Approach to Human Movement Prediction,” describing a real-time GRU pipeline that predicted movement completion from surface EMG with a mean absolute error of 3.49%.

TEACHING & MENTORSHIP

The Salvation Army — Volunteer Tutor & Youth Mentor

Dec 2022 – Present

- Founded and taught a 3D-printing course, tutored students in STEM subjects, and prepared students for competitive high-school entrance examinations.
- Mentored students in exploring career pathways in technology policy, computing, robotics, and game design.

Bioengineering M.A.D. Laboratory I & II — Teaching Assistant

Aug 2025 – May 2026

- Helped develop laboratory curriculum and instructional materials for MATLAB-based bioengineering modeling and design.
- Supported lab instruction through debugging, individual guidance, office hours, and exam reviews.

BOLD Summers — Residential Advisor & STEM Mentor

May – Aug 2024; May – Aug 2025

- Mentored high-school students through individualized research, college, and STEM-career guidance while designing and facilitating academic, professional-development, and community programming.

TECHNICAL SKILLS

Programming & Tools: Python, MATLAB, C#, MicroPython, ROS, MuJoCo, Unity, PyTorch, scikit-learn

Modeling & Control: Machine learning, musculoskeletal modeling, inverse dynamics, quadratic programming, feedback control, signal processing

Experimental Systems: Surface and high-density EMG, OptiTrack motion capture, acoustic signal acquisition, sensor integration, human-subject research

Design & Fabrication: SolidWorks, Onshape, Rhinoceros 3D, circuit design, 3D printing, laser cutting

HONORS & AWARDS

Hugo Otto Wolf Memorial Prize, Penn Engineering

2026

- Received a \$500 prize recognizing exceptionally thorough and original work by a Bioengineering senior.

Career Services Summer Funding Award

2025

- Awarded \$5,000 in competitive funding to support full-time summer research.

Penn Undergraduate Research Mentoring Program (PURM)

2024

- Selected for a \$5,000 faculty-mentored undergraduate research award at the University of Pennsylvania.